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Original Article

Tumour Response in Uveal Melanomas Treated with Proton Beam Therapy

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Abstract

Aims: Post-proton therapy surveillance of uveal melanomas relies on decreased thickness on repeat ultrasound B every 6 months for 2 years and yearly thereafter. Earlier pseudoprogression, a phenomenon described in other tumour types within the first months of irradiation, can also be observed in uveal melanomas and may lead to inappropriate enucleation. The Collaborative Ocular Melanoma Study (COMS) has defined ultrasound criteria to identify tumour progression after brachytherapy. We aimed to determine the reliability of ultrasound as a means to measure tumour height after proton therapy and predict local relapse.

Materials and methods: All 1992–2012 consecutive patients with at least three ultrasound B measurements during follow-up were included.

Results: There were 55 local relapses of 886 patients (6.2%). Ultrasound B reliability was highest at 24 months, with specificity higher than 95% starting at 18 months.

Conclusion: Before 18 months post-proton therapy, the risk of falsely concluding in favour of a relapse exceeds 5% and should prompt repeat measurements 3 and 6 months later but should not prompt enucleation without further clinical assessment.

Key words: Enucleation; proton therapy; regression; tumour response; ultrasound; uveal melanoma

Introduction

Uveal melanoma is the most common primary ocular tumour in adults. It has been treated conservatively since the Collaborative Ocular Melanoma Study (COMS) 20 years ago, based on the demonstration of no detrimental effect on survival or metastasis-free survival with irradiation compared with enucleation [1]. Proton therapy is less widely used worldwide but more appropriate than brachytherapy (with ruthenium or iodine plaques) for thicker/higher tumours (>7 mm) and juxtapapillary tumours (within two optic disk distances, i.e. 3 mm). Increasing availability of proton beam facilities owing to decreasing costs and smaller facility dimensions can be anticipated in the coming years. Increasing numbers of patients may also be anticipated because of earlier detection or because increasing numbers of patients with larger tumours deny upfront enucleation. Moreover, better eye preservation may be anticipated owing to better management of post-therapy complications with retinal surgery and intravitreal injections.

Contrary to all other malignant tumours, a biopsy is not required to make a diagnosis of uveal melanoma. The diagnosis relies on funduscopy and ultrasound. Enucleation or conservative treatments are advocated on the results of these examinations. Fine-needle aspiration is increasingly carried out at diagnosis of uveal melanoma. However, it does not aim to confirm a diagnosis of melanoma but rather to establish a genomic profile of the melanoma. Genomics have been correlated with propensity to develop metastases and are increasingly used for metastatic risk-based strategies (liver imaging, adjuvant systemic treatments). Similar to diagnosis, a biopsy or operative specimen is not requested upon suspicion of relapse. The gold standard for local tumour follow-up is ultrasound based on the

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